

CHAPTER 11: SECURITY & CRYPTOGRAPHY

COMPUTER SECURITY BASICS

- Protecting **confidentiality**, **integrity**, and **availability** (CIA triad) of data and systems.

COMMON THREATS AND ATTACKS

Threat Type	Description	Example
Malware	Malicious software (virus, worm, trojan)	Keylogger
Phishing	Fake emails/websites to steal info	Fake bank email
Denial-of-Service (DoS)	Overloads system to make it unavailable	Flooding requests
Man-in-the-Middle	Attacker intercepts communication	Public Wi-Fi attack
Ransomware	Encrypts user data for ransom	WannaCry

FUNDAMENTALS OF ENCRYPTION

- Encryption:** converts plaintext → ciphertext using a key.
- Decryption:** ciphertext → plaintext using a key.

Types:

- Symmetric encryption:** same key for encryption & decryption (e.g., AES, DES).
- Asymmetric encryption:** public/private key pair (e.g., RSA).

Example:

If plaintext = HELLO

Using key K, ciphertext = XQZZR

Receiver uses K (or private key) to decrypt.

AUTHENTICATION METHODS

- Passwords / PINs** – simplest, but vulnerable to guessing.

- **Biometrics** – fingerprints, face, iris scans.
- **Two-Factor Authentication (2FA)** – combines something you know + something you have.
- **Tokens / Smart Cards** – hardware keys for secure login.

HASHING (MD5, SHA)

- **Hashing:** converts data → fixed-length value (digest).
- Used for verifying data integrity (not reversible).
- **MD5:** 128-bit hash, older, weaker.
- **SHA-2/SHA-3:** stronger, widely used.

Example:

Input = “hello” → SHA-256 = 2cf24dba5fb0a...

CRYPTOGRAPHY APPLICATIONS IN COMPUTERS

- **Data encryption:** files, emails, disks (BitLocker, VeraCrypt).
- **Network security:** HTTPS, VPN, SSH.
- **Authentication:** digital signatures, certificates.
- **Blockchain:** uses hashing + encryption for immutability.

SECURITY COUNTERMEASURES

Tool	Purpose
Firewalls	Filter incoming/outgoing network traffic.
IDS/IPS	Detect/prevent intrusions.
Antivirus software	Identify & remove malware.
Access controls	Restrict users' privileges.
Regular updates	Patch vulnerabilities.

CASE STUDY — HTTPS (ENCRYPTED COMMUNICATION)

- HTTPS = HTTP + SSL/TLS encryption.
- Browser and server exchange digital certificates.
- Session key established → encrypts all traffic.
- Ensures **confidentiality**, **integrity**, and **authentication** between client & server.

QUICK RECAP & STUDY QUESTIONS

- What are the three pillars of security (CIA triad)?
- Difference between symmetric and asymmetric encryption?
- What does hashing ensure?
- Give two countermeasures against malware.
- Why is HTTPS more secure than HTTP?